

Web Site Heuristics

This document lists:

- A wide range of heuristics related to usability and web site quality that AI can likely detect
- An information hierarchy for organizing these heuristics, useful when presenting an analysis report to end-users
- Marketing, to help explain to folks what we offer

This hierarchy drew on heuristics from a number of sources, including:

- [FuguUX Ideas for New Analyzers](#)
- [FuguUX Heuristics to Try with LLMs](#)
- [FuguUX Example Bad Interfaces](#) (lots of examples here)
- [Ziyan's initial hierarchy](#)
- Jason's Design Patterns book
- Don't Make Me Think book ([see our notes here](#))

Also see404

- Should have most important content Above the Fold
- Should make good use of negative Space
- Should have a stable layout (rather than one that changes as the page continues to load or when users interact with it)
- Should have emphasis
 - Should have most important content be more prominent than everything else
- Should make good use of Color
 - Should conform to standard Color Theory
 - Should have Color Harmony - even if contrast is acceptable, colors should not clash
- Should have contextually appropriate images that reinforce key themes
 - Should not have pixelated images
 - Should not resize images in browser due to graininess or inefficiency or general weirdness
 - Should aim to have images match golden ratio
 - Should minimize file size of images
- Should have appropriate multimedia
 - Should not have very large videos autoplay
 - Should not autoplay videos with sound on
 - Include captions on videos for accessibility
- Should have 3-5 macro visual elements on the page that are coherent and grouped well ([see here for more info](#))
 - Should not have web elements overlap on top of each other
 - Should use floating divs appropriately
 - Should not have floating divs cover important content
 - Should handle scrolling with floating navbars correctly
- Should have good text readability
 - Do not use pure black to reduce eyestrain

- Should have appropriate font size (not too small or too big)
- Should have text that has good contrast against background
- Should have minimal use of center-aligned text
- Should have appropriate line width and line height (not too narrow, not too wide, line height should be approx. 1.4-1.6 times the font size)
- Should have margins, so that does not go all the way from one edge of browser window to the other edge
- Should have appropriate line spacing (not too narrow or wide)
- Should have consistency in text alignment
- Should minimize use of center-aligned text
- Should have minimal distractions
 - Should not have a distracting background image
 - Should not have too many ads
 - Should not have too much animation
 - Should not have too much italicized, bolded, or underlined text
- Should not have a single word span more than one line
- Should minimize use of ALL UPPERCASE (ok for headers, labels, acronyms)
- Should use typography appropriately
 - Should use web-friendly typefaces
 - Should use readable fonts (e.g. not too thin)
 - Should not have ugly fonts (maybe have set common ones like Comic Sans, Papyrus, etc. as the criteria for ugly fonts)
 - Should not have too many different fonts (typefaces + sizes)
 - Should not have fonts that are too similar (e.g. one is bold and another is larger or two typefaces from same category)
- Should not use default underline
- Should only use at most one of underline, bold, and italics
- Should use curved quotes and apostrophes rather than flat ones
- Should have a favicon (for browser tabs, bookmarking, history)
 - Favicon must be a multiple of a 48px square
 - Should be an image or graphic that shows up well at a tiny size
 - Google will replace inappropriate favicon content with the default image

2. Site Content

- Text Content
 - Should have readability level appropriate for audience
 - Should have understandable text for target audience
 - Should minimize use of jargon (or appropriate for target audience)
 - Should not have typos
 - Should not have HTML / CSS / JS / JSON leaks in the text
 - Should not have currency errors (e.g. \$123.1)

- Should not have weird characters in text (encoding error or copy-paste error)
- Should not have paragraphs that are too long
- Should not have instructions that are too long, if the text is instructions
- Should not have empty bullets
- (For task-oriented sites) Should avoid meaningless text content
 - Ex. Welcome to...
- Page content
 - Header
 - Should have a clear brand
 - Should have clicking on top left logo take you to home page (convention)
 - Should have standard items common at top for the genre (e.g. login, navbar, search), follows convention
 - Should not have too much vertical whitespace in header so that it inadvertently pushes useful content below the fold
 - Footer
 - Should have standard items common at bottom for the genre (e.g. link to privacy policy, contact us, site map, etc, see my design patterns book too)
 - Should have common secondary links
 - Should have any social networks
 - Should have site map
 - Should not have broken 404 images
 - Should not overwhelm readers with too much information
 - Content module
 - Should not have empty content module
 - Lists (of links, content, images, etc)
 - Should be sorted in some meaningful way (e.g. hierarchically, task-based, alphabetically, chronologically, by popularity)
 - Should use bullets and not hyphens or dashes for bullets
 - Should be grouped together in a cohesive manner (possibly with headers) if there are many items in the list
 - Paragraphs (for task-oriented sites vs reading ones)
 - Should have multiple headers to help with skimming
 - Should have short paragraphs
 - Use bulleted lists (and include a small amount of space between items in the list)
 - Highlight key terms (though not too much, too much bolding or highlighting can be distracting)
 - Instructions
 - Should have instructions be minimal and self-explanatory
- Internationalization
 - URL should be set up for the country
 - Hreflang tag should be set up for the country
 - If using a CMS, should be one that accounts for different languages

- Should accommodate text expansion and contraction to adapt to new languages
- Should design for right-to-left languages as well if appropriate
- Should localize currencies, time, date formats, etc.
- Credibility
 - Certain types of URLs lend more credibility (.org, .edu, .gov, .com, etc.)
 - HTTPS is more secure than HTTP
 - Should have terms of service and privacy policy
 - Limit stock images
 - Limit advertisements
 - If the website is based on a brick and mortar place, try to bring elements of that onto the site (such as physical address, employee photos)
 - Should be easy to find contact information for the company running the website
 - Cite credentials of sources of information
 - Having third party credentials helps
 - Don't make amateur errors like typos or broken links
- Common pages
 - Homepage
 - Should have a clear value proposition
 - Should have strong first impression
 - Content page
 - Should have clear headline and blurbs to help with skimming
 - Should have printable pages that strip out navigation, ads, and other ancillary content
 - About Us
 - Should have a physical address
 - Should have a support phone number or email
 - 404 not found page
 - Should have a helpful 404 not found page
 - Privacy Policy
 - Account management
- Common tasks
 - Find contact information
 - Create account and login
 - Check notifications
 - Add item to cart and checkout
 - Search for page or information
 -

3. Performance (speed, size, platforms, compatibility)

- Should have fast load times
 - Should have fast load time for page content

- Should minify JS Code for high traffic sites
- Should not have large image sizes (except for image galleries)
- Should not have large video sizes
- Should not have web site load unused JS libraries
- Should not load too many big JS libraries
- Should not take up too much browser memory
- Should not have browser memory usage increase over time even without active usage
- Interaction performance
 - Should have fast loading animation times
 - Should not have lag when doing user input (mouse, keyboard, touch)
- Should have good performance over Cellular Networks / poor networks
- SEO Optimization
 - Should have good metadata
 - Should have good metatags and descriptions
 - Should have good keywords
 - Should have an overall sitemap for search engines
 - Should have unique page titles for all pages
 - Should have good page titles for all page titles that accurately reflect the content on the page
 - Should not have invisible text (text that is similar or same color as background)
- Lighthouse metrics
 - (LCP) Largest Contentful Paint, measures perceived load speed and marks the point in the page load timeline when the page's main content has likely loaded <https://blog.chromium.org/2020/05/introducing-web-vitals-essential-metrics.html>
 - (FID) First Input delay < 100msec, measures responsiveness and quantifies the experience users feel when trying to first interact with the page <https://blog.chromium.org/2020/05/introducing-web-vitals-essential-metrics.html> and https://developer.mozilla.org/en-US/docs/Glossary/First_input_delay
 - (TTFB) Time to First Byte, refers to the time between the browser requesting a page and when it receives the first byte of information from the server. This time includes DNS lookup and establishing the connection using a TCP handshake and SSL handshake if the request is made over https. TTFB is the time it takes between the start of the request and the start of the response, in milliseconds <https://developers.google.com/web/fundamentals/performance/speed-tools> and https://developer.mozilla.org/en-US/docs/Glossary/time_to_first_byte
 - (FCP) First Contentful Paint, measures how long it takes the browser to render the first piece of DOM content after a user navigates to your page. Images, non-white <canvas> elements, and SVGs on your page are considered DOM content; anything inside an iframe isn't included. <https://developers.google.com/web/fundamentals/performance/speed-tools> and <https://web.dev/first-contentful-paint/>
 - (FI / FCI) First Interactive, also known as first CPU idle, is a non-standard web performance metric that measures when the user's window is quiet enough to

handle user input, or what is termed as minimally interactive.

https://developer.mozilla.org/en-US/docs/Glossary/First_interactive

- (FMP) First Meaningful Paint, is the paint after which the biggest above-the-fold layout change has happened and web fonts have loaded. It is when the answer to "Is it useful?" becomes "yes", upon first meaningful paint completion.
https://developer.mozilla.org/en-US/docs/Glossary/first_meaningful_paint
- (FP) First Paint, part of the Paint Timing API, is the time between navigation and when the browser renders the first pixels to the screen, rendering anything that is visually different from what was on the screen prior to navigation. It answers the question "Is it happening?"
https://developer.mozilla.org/en-US/docs/Glossary/First_paint

4. Interaction Flow / Human-Centered Design

- Accessibility
 - Should comply with WAI Criteria
 - Should comply with WCAG Criteria
 - Should comply with ARIA
 - Should support standard Keyboard Navigation
 - Should have a default keyboard focus make sense
 - Should have tab ordering be logical and make sense (left to right, top to bottom)
 - Should have arrow keys work when conceptually they should (e.g. image gallery, item scrolling, etc)
 - Should have escape key close modal dialog boxes
 - Should not have text input performance be laggy
 - Should have fast keyboard text entry be correct (e.g. too much JS can make things laggy and incorrect, I've seen Don't become ont'D)
 - Should have Alt Text for all images
 - Should have good ALT text that matches the image
 - Should have page be understandable for people with various kinds of color-blindness or color deficiencies
 - Should have redundant cues to help with navigation
 - Should not have fixed height buttons
 - Should not have fixed height text boxes with overflow: hidden

Navigation

- Should have highly visible navigation
- Should have navigation be persistent across site (except for checkouts and other process flows)
 - Should have site logo
 - Should have utilities (e.g. sign in, settings, profile, contact)

- Should have search bar
 - Should have search bar conform to conventions (e.g. button is Go or Search or magnifying glass)
 - Should have primary sections of the site (navbar)
- Should have strong consistency of navigation elements across site (look and feel, terms used)
- Should have most important actions be prominent
 - Should have clear call to actions (buttons or links)
 - Should not have multiple calls to actions on same page
 - Should have most common tasks be more visible
 - Should have most important flows take fewer steps to complete
 - Should have primary actions differ visually from secondary actions
- Should have clear page number if content is paginated across multiple pages
- Should have most important navigation be visible (above the fold, no vertical or horizontal scrolling)
- Should not overwhelm users with too many choices
- Should not have critical and irreversible actions too close to common actions
- Navigation Bar and Menus
 - Should have clear and useful category names and terms in navigation bar / good information scent
 - Should not have too many menu items in the navbar
 - Should not have just one item in menu
 - Should have consistent names for items in navbar across site
 - Should have distinct names for items in navbar
 - Should have menu items be sufficiently spaced
 - Should not have navigation bar fold over to next line if not at full width
 - Should have entire relevant area of navbar be clickable (e.g. the row or the box, rather than just the text label) (Fitts' Law)
 - Should have navbar elements be coherent and grouped together well
 - Should use ... ellipses in submenus if clicking on it goes to a dialog box to continue
 - Should not have menus that are too sensitive on mouseover, leading to accidentally triggering other menus (Fitts' Law)
- Search
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- Modal dialog boxes
 - Should have modal dialogs appear on screen (vs modal dialog box appearing off screen, off view, or behind browser window, making it appear that things have locked up)
 - Should have keyboard Escape key close the modal
 - Should have close button be clearly inside of the modal dialog box (e.g. some ads have the X far outside of the ad, making it hard to tell how to close)

- Should have context-specific labels for buttons vs generic ones, e.g. “Move to trash” vs lots of text explaining what will happen and “Ok” or “No” or “Yes”
- Should have keyboard focus on the modal dialog when it is opened
- Forms
 - Should have forms be consistent with each other across site
 - Should have common flows conform with common design patterns
 - Login flow
 - Should not confuse users with login vs account creation
 - Should support recovery for forgotten passwords
 - Account creation flow
 - Should show password requirements before (not after trying to create a password)
 - Shopping cart flow
 - Should have email notification after completion of flow
 - Text input
 - Should only require minimal text input
 - Should not have to fill in the same information in a form or a flow more than once (e.g. mailing street address and street address for credit card)
 - Should have default keyboard focus if there is only one input in the main content area (e.g. two factor authentication input)
 - Should not have ambiguous formats (e.g. dates and times due to internationalization)
 - Should allow <Enter> to go to next field (or submit form if on the last field in a form)
 - Should have empty states and labels be clear and friendly
 - Should have placeholder text where needed, clearly indicating what should be put into the form
 - Names
 - Should support really long names
 - Should support really short names
 - Should test for last name NULL
 - Should test for names with space or dash in it
 - Should test for name with apostrophe in it
 - Form Layout
 - Should have labels proximally close to their corresponding input field
 - Should not have form elements run too wide (especially entire width of page)
 - Should break up into multiple sections or pages if the form is long
 - Should not have form elements be spaced too close to each other
 - Form Errors
 - Should have good error prevention (example inputs, what format needed, what's required vs not, etc)

- Should have clear phrasings that make clear what information is needed in a field
 - Should have clear markings that make clear what fields are required
 - Should use proper HTML input field types to help structure the data and prevent errors
 - Should avoid overly aggressive form validation
 - Should avoid overly aggressive form validation to prevent SQL injection and XSS attacks
- Should have good error recovery
 - Should not have change blindness (where page appears too similar to before after submitting)
 - Should clearly mark where the error is, using redundant cues to avoid change blindness and so that people with color deficiency will notice it
 - Should have error message is in plain language
 - Should precisely indicate the problem with minimal jargon
 - Should constructively suggest a solution
- Should correctly save form data on network error or on back button
- Submit button / Confirmation
 - Should have clear and high-visibility Submit (or Continue or Next) button
 - Should not disable submit button (better to let users know what's wrong or what's missing rather than guessing why submit button is disabled)
 - Should have a confirmation page after submitting data correctly (and possibly a confirmation email as well, depending on the form)
 - Should not have a reset form button, as it can be clicked on by accident
- Widgets
 - Should have widgets be consistent with each other across site
 - Should have clear affordances, feedback, and feedforward
 - Should be greyed out if inactive
 - Should explain why something is inactive (e.g. tooltip or adjacent text)
 - Should not be greyed out if active
 - Should have a mouseover hover effect (e.g. changes cursor to look clickable or changes the button appearance)
 - Should have tooltip if functionality is not immediately obvious
 - Should have tooltip be a good description of functionality
 - Should have slight delay to tooltip (vs showing immediately on hover)
 - Should appear clickable (ie minimal flat design)
 - Should have the correct z-axis order
 - Should minimize use of horizontal scrolling
 - Should have widgets conform with common patterns and norms
 - Links
 - Should have clear terms in links / Good information scent

- Should not have non-link text look like a link (avoids false positive)
- Should have link text look like a link (avoids false negative)
- Should have link color be visually distinct from non-link text
- Should have link color that has good contrast against background
- Should have visited link color that has good contrast against background
- Should have good visited link color that draws the eye less than unvisited links
- Should have links serve as links rather than actions (e.g. submit)
- Should not have 404 dead links
- Should not have empty href
- Should have link text that is distinct from other links that go to different parts of the site
- Should have logical groupings of hyperlinks if there are many of them (beyond 7 +/- 2)
- Should separate unsafe links (e.g. logout, delete) from safer links
- Should act like a link
- Button
 - Should have button sizes that are similar
 - Should have a clear button hierarchy (if applicable)
 - Should have a text label with a clear and unambiguous action
 - Should minimize use of buttons that only have icons
 - Should have a label that is action oriented (verb rather than noun)
 - Should not have a label that is too long (makes button fat or wide)
 - Should have button show entire label text
 - Should have entire button area be clickable (Fitts' Law)
 - Should have appropriate target size
 - Should not have non-buttons look like a button (avoids false positive)
 - Should appear to be buttons (avoids false negative)
 - Should do actions (e.g. submit, cancel)
 - Should have images that are appropriate and semantically match the text label
 - Should have button text that has sufficient contrast against the button background
 - Should have redundancy to prevent accidents for dangerous actions
 - Should have + or some kind of chevron if clicking on the button leads to more choices
 - Should have clear indicator of which button is selected if there are grouped buttons
- Sliders
- Radio Buttons
 - Should have radio buttons be select exactly one (vs checkboxes)

- Should have button + associated label be clickable via <label>
- Checkboxes
 - Should have checkboxes be select as many as wanted (vs radio buttons)
 - Should have button + associated label be clickable via <label>
- Date/Time Pickers
- Icons
 - Should have consistent icon images (rather than from multiple collections)
 - Should have good color contrast ratio with icons
 - Should have tooltips
 - Should have text label if space allows
 - Should have optically centered icons rather than geometrically centered
 - Should have icon with good color contrast ratio
- Dividers
- Email Inputs
- Dropdown Box
 - Should not have duplicate items
 - Should not be empty
 - Should be sorted in some meaningful way (e.g. hierarchically, task-based, alphabetically, chronologically, by popularity)
 - Should support typing for jumping to result
 - Should have a scrollbar if there are more items
 - Should have arrow keys work correctly to go up and down
 - Should differentiate between 12noon and 12midnight for time input widgets
 - Should have "Other" as last option if it exists (rather than in the middle)
 - Should not have excessively long dropdown fields
- List Box
 - Same as Dropdown Box
- Search Bar
 - Should keep search term in the search form field on the search results page
 - Should have search bar be readily apparent if it exists
 - Should have sample text to help users understand that it is a search bar (e.g. "search for xxx")
 - Also see Dropdown box, the search bar auto-complete should have similar properties
 - Should have tags or an auto-suggest mechanism to help user understand what they can search
 - Should have filters and ways to customize filters
- Text Input Fields

- Should not have grey text (as that makes it appear disabled)
 - Should be appropriate size based on expected input size
 - Should allow HTML input autocomplete for basic data types
 - Should enable HTML input autofocus if no other competing elements
 - Should not have laggy text input
- Text Areas
 - Should not have grey text (as that makes it appear disabled)
 - Should be appropriate size based on expected input size
 - Should have TextArea be resizable (i.e. should not use `resize:none`)
 - Should not have laggy text input
- File Uploads
- Range Inputs
- Scrollbar
 - Should have scrollbar if content is scrollable
 - Should minimize use of scrollbars in divs (double scrollbars)
- On/Off Toggle
 - Should minimize use of On/Off Toggle buttons due to potential confusion
- Divs
- Tables
 - Should have sticky table headers if a long table
 - Should have long tables be sortable by headers
 - Should have consistent table row heights
 - Should not have text in table be center aligned
 - Should not have columns be too narrow for the text contained
- Direct Website Entry
 - Should have web page make sense when entering via a search engine
- Personalization
 - should provide data collection information
 - should give customized options for user autonomy
- Notifications and Alerts
 - should be limited to reduce alert fatigue
 - should have clear instructions
 - should be able to timeout automatically
 - should allow user to dismiss or check them off
 - Should have alert messages be consistent and follow the rules of a snack bar, toast, and banner

5. Standards and Compliance

- Standard Browser Functionality
 - Should not break standard browser functions (back button, keyboard shortcuts, etc)
 - Should have Back button work correctly
 - Should correctly update history state so that Back button works correctly
 - Should have keyboard Ctrl-A selects all content in the focus
 - Should have keyboard Ctrl-X cut
 - Should have keyboard Ctrl-C copy
 - Should have keyboard Ctrl-V paste
 - Should have keyboard Ctrl-F find
 - Should have keyboard Ctrl-N open new browser window
 - Should have keyboard Ctrl-O open
 - Should have keyboard Ctrl-P print
 - Should have keyboard Ctrl-R reload
 - Should have keyboard Ctrl-S save
 - Should have keyboard Ctrl-T open new tab
 - Should have keyboard Home go to top of page
 - Should have keyboard End go to bottom of page
 - Should have keyboard PageUp go up a page
 - Should have keyboard PageDown go down a page
 - Should have keyboard arrow keys go up and down (if focus is not in a text area)
 - Should have keyboard Alt-Left go back a page
 - Should have keyboard Alt-Right go forward a page
 - Should let mouse dragging highlight text
 - Should not remove browser scrollbar (see scrolljacking)
 - Should have right mouse button activate popup menu
 - Should support basic zooming and page should still appear mostly correct (also see Responsive Design)
- Privacy
 - Privacy Policies
 - should be in regular language, not jargon
 - should be formatted for easy reading
 - make it easy for the user to choose their online tracking, keep the label transparent
 - explain why the data is being used and shared and whom it is shared with
 - give the user contact info if they have questions or concerns
 - Minimal Data Collection
 - Data Protection
 - User Consent

- Security
 - Should use https for logins
 - Should use https when transmitting sensitive data (PII, credit card info, etc)
 - Should avoid metadata for sensitive content (e.g. JPG EXIF, PDF metadata, docs and xlsx metadata)
 - Should avoid URL Mucking (e.g. modifying the URL to get access to things one shouldn't have access to)
 - Should avoid having directory of employees online (phishing attack potential)
 - Should not have username and password publicly visible on a page (e.g. in a README file)
 - Should not have automatic logout times that are too fast
 - Should not have libraries that are out of date and have known security issues
 - Should not allow XSS security errors
 - Should not have bad code copy-pasted from StackOverflow
 - Should aim to minimize the display sensitive data in the page due to caching (e.g. SSN, bank account number, etc)
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- Domain name and URLs
 - Should aim to have readable URLs
 - Should have web page be able to load without the www prefix
 - Should not have domain be expiring soon
 - Should have http:// and https:// version of domain work
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6. Web Development Foundations

- HTML
 - Should be valid HTML
 - Common bugs
 - Should not have more than one item with autofocus set
 - Should have unique ids for each web element in a page
 - Should use <H1> <H2> ... <H6> to maintain hierarchy correctly if used
 - Should use a single <H1> for the main title
 - Should use heading tags hierarchically
 - Should not skipping heading levels
 - Should use <H1> for Page Titles, Not Logos or Banners
 - Should use semantics of header tags correctly
 - Doctype should be declared
 - Use HTML to show hierarchy
 - Close tags
- CSS
 - Should be valid CSS
 - CSS Layout Methods
 - Flexbox should follow the natural flow of the content

- use proper flex properties and alignment
 - Minimize flex items in a single container, avoid complicated nesting
 - Use breakpoints for flexboxes
- Selectors
 - Names of classes and IDs should be appropriate and descriptive
 - IDs should be used sparingly
 - Use appropriate naming conventions (such as BEM)
 - Avoid using universal and overly specific selectors if possible
- Box Model
 - There should be enough padding around clickable areas especially for touch screens
 - Avoid overlap on items
- Javascript
 - Should be valid JS
 - Should minimize number of error messages to console
 - DOM Tree
 - DOM tree should not be overly nested, limit number of elements in a DOM
 - Use semantic elements to convey meaning for screen readers
 - structure should be logical
 - Keep HTML, CSS, and JS separate
 - Make sure not using legacy JS
 - Event Listeners
 - minimize number of event listeners on a page to improve performance
 - delete unused listeners
 - use standard event types
 - Animations
 - Avoid complex animations
 - Should provide clear feedback and affordances
 - Enable options to allow motion-sensitive users to reduce animations

7. Responsive Design

- Should adapt to different screen sizes and still look good
 - Should look good for smartphones
 - Should look good for tablets
 - Should look good for desktops
- Should adapt to landscape vs. horizontal
- Browser compatibility

- Should work properly for Chrome, Edge, Safari, Firefox desktop web browsers
- Should work properly for smartphone web browsers
- Should work properly for tablet web browsers
- Mobile and Tablet
- Gestures
 - Should always allow pinch zoom gesture
 - Should always allow standard panning gestures (left, right, up, down)
- Should be touch-friendly
 - Should have minimum hit sizes (see Material Design and Apple Human Interface Guidelines)
 - Should have minimum spacing between clickable items (see Material Design and Apple Human Interface Guidelines)

8. Delight (holistically putting it all together)

- Should not have too many things competing for a person's attention
 - Ex. Too much animation
 - Ex. Lots of invitations to buy
 - Ex. Lots of exclamation points
 - Ex. Lots of different typefaces
 - Ex. Lots of bright colors
 - Ex. Automated slideshows
 - Ex. Pop-ups
- Should not offer too many choices for the same thing at the same time
 - Ex. logins
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What usability problem (specific)

What usability problem (general principles and patterns)

Where (site wide or specific page)

Customer Feedback

Microinteractions

Individualized Content

Dynamic Content

Loading animations

Inclusivity

Search Functionality

Tags
Exact vs. Fuzzy Search
Media Scaling
Filters
HTML Inputs
CSS Layout Methods
Animations
Browser Interactions
Media Scaling
Graphic Elements
Language Options
Videos
Live Feed
Notifications
Styling
Placement Hierarchy
Backups
User Authentication
Structural Elements
DOM Tree
Website Structure
Motion Graphics

Human-Centered Design
Performance and Security
Web Development Foundations

Email heuristics

Also See

 Examples of Bad Interfaces That Should be Detectable

Use Cases

Check email before sending

Check bulk email before sending

Bugs (To Be Added to Heuristics Below)

- Hey [First Name], Try this LinkedIn outreach that works
- Sat Mar 11 (when it's Sun Mar 11)
- Not including timezones when discussing times with people in other timezones
- Checking for daylight savings if the conversation is before then

Email heuristics

Bias

Anger

Fake email response

Ambiguity (e.g. missing a date for meeting), increases number of turnarounds needed

Too Long

Subject line

Good subject line

Incorrect Email addresses

Email contains an email address that is incorrect (or common typo)

Too Many Email Addresses

Risky, too many people, leads to "take me off your mailing list" errors

Incorrect URL

Email contains a URL that goes to 404 not found

Email contains a URL that is incorrect (or common typo)

Email contains a URL that is unsafe (known phishing site)

Quoted printable encoding problem where = becomes =3D

Text Content

Images

Email image with no ALT tag

Text Layout

HTML email text too wide, can't read it without scrolling

Bugs

Dear {First Name}

Image doesn't load

Subject line typos

Bad link

Leaked Code in email

Leaked Code in email

Email Navigation

Looks like a button but no link / isn't clickable

Button area looks clickable, but only text is

Email Forwarding

If email is forwarded, still readable

Email Reply-To

For mailing lists

Email Safety

Make sure that links don't lead to auto-actions (since virus and phishing scanning), include extra confirmation on web page

Scheduling

Missing basic things like time zone

Mismatched date (Tue Jan1 when it's Wed Jan1)

Meeting needs date

Meeting needs time

HTML

HTML email should also contain text only

Footer

Unsubscribe option if bulk email

Anti-Phishing

Getting people to avoid bad habits, like clicking on links to banks or other sensitive sites
Link text should match URLs

Copy-Paste

Problem: I copy email notifications from GitHub (typically CVE alerts) and paste into Slack, and it looks terrible

Related: copy-paste 2FA sent to email into the web site

Unsubscribe

Every bulk email should have unsubscribe at bottom footer

Mobile Heuristics

Mobile Heuristics

These are drawn (in part) from the Web Site Heuristics tab of this same document + Microsoft CoPilot

◆ Layout, Responsiveness & Visual Stability

- Responsive layout adapts cleanly — No horizontal scrolling, no clipped text, no overlapping elements.
- Tap targets are large enough — Buttons/links have at least ~44px touch area and adequate spacing.
- Text is readable without zooming — Proper font sizes, line spacing, and contrast.
- No layout shift during load — Elements don't jump around as ads/images load (CLS issues).
- Images scale correctly — No stretched, pixelated, or oversized images.

◆ Navigation & Interaction

- Navigation is discoverable and reachable — Hamburger menu or bottom nav is easy to find and use.
- Scrolling is smooth — No jank, lag, or scroll hijacking.
- Back button works predictably — No traps, unexpected reloads, or modal overlays blocking navigation.
- Forms are mobile-friendly — Proper input types (email, number), auto-focus, and no keyboard overlap.

◆ Content & Readability

- Content hierarchy is clear — Headings, spacing, and grouping make scanning easy.
- No intrusive pop-ups — Consent banners, promos, or sign-ups don't block the screen or require precision taps.
- Media loads efficiently — Videos/images don't stall or freeze the page.

◆ Performance & Technical Quality

- Fast initial load — Page becomes usable within a couple of seconds on mobile data.
- Optimized assets — Compressed images, minimized scripts, lazy loading where appropriate.
- Minimal blocking scripts — No long pauses before interaction is possible.

- Works well on low-end devices — Acceptable performance even with limited CPU/RAM.

Accessibility Heuristics

Accessibility Heuristics

These are drawn from the Web Site Heuristics tab of this same document + Microsoft CoPilot

- 21. Should comply with WAI Criteria
- 22. Should comply with WCAG Criteria
- 23. Should comply with ARIA

ARIA labels are present where needed — Buttons, icons, and dynamic components.

No incorrect roles, no redundant ARIA, and no ARIA hiding visible content.

- 24. **Keyboard navigation works** — All interactive elements are reachable via Tab.
- 25. Should have tab ordering be logical and make sense (left to right, top to bottom)
- 26. Should have a default keyboard focus make sense
- 27. Should have arrow keys work when conceptually they should (e.g. image gallery, item scrolling, etc)
- 28. Should have escape key close modal dialog boxes
- 29. **Headings follow a logical hierarchy** — No skipped levels (e.g., jumping from H1 to H4), and headings reflect structure, not styling.
- 30. **Text is readable** — Adequate contrast, font size, spacing, and no clipped or truncated text.
- 31. **Links are descriptive** — Avoid “click here” or “read more” without context; link purpose should be clear.
- 32. **Skip-to-content link exists and works** — Allows keyboard users to bypass navigation.
- 33. **Interactive elements use proper HTML elements** — Buttons are `<button>`, links are `<a>`, not `<div>` or `<span>` with click handlers.
- 34. **Page title is descriptive** — Each page has a unique, meaningful `<title>`.
- 35. **Alt text exists for images** — Especially for meaningful or functional images. Decorative images use empty alt (`alt=""`), functional images describe their purpose.
- 36. **Media has captions or transcripts** — Videos include captions; audio content has transcripts.
- 37. **Buttons have accessible names** — Icons (hamburger, search, close) have labels readable by assistive tech.
- 38. **Dynamic content is announced** — Modals, alerts, and validation messages are announced via ARIA roles or live regions.

39. Responsive layout does not break accessibility — No overlapping text, no hidden content, no unreachable controls on small screens.

40. Visible focus indicators — Clear outlines or highlights on focused elements.

41. Hover/focus states are distinct — Interactive elements show clear visual changes on hover and focus.

42. Modal dialogs trap focus properly — Focus moves into the modal, stays inside, and returns to the trigger when closed.

43. Color contrast is sufficient — Text meets WCAG contrast ratios. Text and UI elements meet contrast guidelines; nothing relies on color alone to convey meaning.

44. Form labels are correctly associated — No placeholder-only labeling

45. Forms have proper labels — Every input has a programmatically associated label; placeholders are not used as labels.

46. Error messages are clear and linked to fields — Errors appear near the field, use text (not just color), and are announced to assistive tech.

- Should not have fixed height buttons
- Should not have fixed height text boxes with overflow: hidden

Quality Assurance Heuristics

Quality Assurance Heuristics

These are drawn from the Web Site Heuristics tab of this same document + Microsoft CoPilot

A. Functional Quality (Core Behavior)

1. **No console errors** — No red errors in DevTools on load or interaction.
2. **All links work** — No 404s, no dead external links, no unexpected redirects.
3. **Forms submit correctly** — Validations trigger, errors display, success states work.
4. **Buttons and controls respond** — No dead UI elements or unresponsive clicks.
5. **Search works** — Returns results, handles empty queries or invalid queries gracefully, and shows no broken states.
6. **Media loads properly** — Images, icons, and videos appear without broken placeholders.
7. **Downloads function** — Files download with correct names and formats.
8. **Critical flows complete** — Checkout, login, signup, and other key journeys work end-to-end.

B. Layout & Visual Quality

9. **No overlapping or clipped elements** — Text and UI components stay within boundaries and do not overlap..
10. **Consistent spacing and alignment** — No random padding or misaligned blocks or unintended whitespace..
11. **Typography is consistent** — Font sizes, weights, and styles match the design system.
12. **Responsive layout works** — Breakpoints adapt cleanly; no horizontal scrolling.
13. **Images scale correctly** — No stretching, pixelation, or wrong aspect ratios.
14. **No layout shift during load** — Avoids content jumping (CLS issues).
15. **Consistent visual design** — Colors, typography, spacing, and component styles match the design system.
16. **Media loads and plays correctly** — Images, videos, and icons appear at the right resolution and format.

C. Usability & Interaction Quality

15. **Navigation is intuitive** — Menus are discoverable, predictable, and consistent.
16. **Navigation is consistent and functional** — Menus, breadcrumbs, and internal links behave predictably across pages.
17. **Interactive states exist** — Hover, focus, active states are visible and meaningful.

- 18. **Pop-ups are dismissible** — Pop-ups, cookie banners, and modals are dismissible and non-disruptive. No traps, no tiny close buttons, no blocked content.
- 19. **Scrolling is smooth** — No jank, scroll hijacking, or infinite-scroll glitches.
- 20. **Back button works** — No loops, no unexpected reloads, no blocked navigation.
- 21. **Content hierarchy is clear** — Headings, grouping, and spacing support scanning.
- 22. **All interactive elements respond** — Buttons, toggles, accordions, and tabs react immediately and correctly.
- 23. **Forms validate properly** — Required fields, error messages, and success states work as expected.
- 24. **No unexpected auto-playing media** — Audio/video does not start without user action unless explicitly intended.
- 25. **Error states are clear and helpful** — 404 pages, form errors, and system messages guide the user toward recovery.
- 26. **Content is up to date** — No outdated dates, expired promotions, or obviously stale information.

E. Performance & Technical Quality

- 27. **Page loads quickly** — First meaningful interaction within a few seconds.
- 28. **Assets are optimized** — Compressed images, minified JS/CSS, lazy loading.
- 29. **No excessive script errors or warnings** — No repeated retries or failed network calls.
- 30. **Caching works** — Static assets are cached; no unnecessary reloads.
- 31. **No memory leaks** — Page doesn't slow down after extended use or navigation.

F. Trust, Security & Content Quality

- 32. **HTTPS everywhere** — No mixed content warnings.
- 33. **No suspicious redirects or injected ads** — Clean, predictable behavior.
- 34. **Error pages are helpful** — Custom 404/500 pages with recovery options.
- 35. **Content is up to date** — No stale dates, expired promotions, or lorem ipsum.
- 36. **Privacy and cookie banners behave correctly** — Respect choices, don't reappear endlessly.

First Impressions Heuristics

First Impressions Heuristics

These are drawn from Microsoft CoPilot and Mike's suggestion of McKinsey Design Index

- **Brand alignment:** aligns with our company's brand vision
- **First impressions:** creates a positive first impression
- **Memorability:** creates at least one "wow" moment for the user

Heuristics

A. First Impressions (Immediate Impact)

Clarity within 3 seconds

1. The user instantly understands what the site is about and who it's for.

Strong visual hierarchy

2. The eye is guided naturally — clear primary message, supporting visuals, and obvious next steps.

Clean, uncluttered layout

3. No overwhelming text, no chaotic spacing, no visual noise.

High-quality hero section

4. The hero image/video feels intentional, crisp, and emotionally aligned with the brand.

Professional typography

5. Fonts feel modern, readable, and consistent — no mismatched styles.

Immediate trust signals

6. Subtle cues like quality visuals, polished UI, and stable layout create instant credibility.

B. Brand Alignment (Identity & Consistency)

Consistent color palette

7. Colors match brand guidelines and reinforce the intended mood (e.g., energetic, calm, premium).

Tone of voice matches brand personality

8. Headlines, microcopy, and CTAs reflect the brand's character (playful, authoritative, minimalist, etc.).

Visual style coherence

9. Icons, illustrations, photography, and UI elements share a unified aesthetic.

Brand story is visible

10. The site communicates values, mission, or differentiators without requiring deep digging.

Logo usage is correct

11. Proper sizing, spacing, and placement — no distortion or low-resolution assets.

Consistent interaction patterns

12. Buttons, animations, and hover states behave in a way that feels “on brand.”

C. Wow Factor (Delight & Emotional Resonance)

Thoughtful micro-interactions

13. Subtle animations, hover effects, or transitions that feel smooth and intentional.

High-quality imagery or video

14. Visuals feel premium, emotionally engaging, and relevant — not generic stock.

Polished motion design

15. Animations enhance the experience without slowing it down or feeling gimmicky.

Memorable signature element

16. A unique visual, interaction, or storytelling moment that sets the site apart.

Modern design patterns

17. Use of contemporary layouts, spacing, and visual trends without overdoing them.

Emotional resonance

18. The site evokes a feeling — excitement, trust, curiosity, calm — aligned with the brand's intent.

D. Credibility & Professionalism (Supporting the Impression)

No visual glitches

19. No broken images, misaligned elements, or layout shifts during load.

Fast initial load

20. A slow site kills the wow factor instantly.

Readable and polished copy

21. No typos, awkward phrasing, or filler text.

Clear, confident CTAs

22. Buttons feel purposeful and inviting, not generic or confusing.

Consistent spacing and rhythm

23. Vertical rhythm and whitespace feel intentional and balanced.